

School of Information Technology

Department of Computer Science



COS720 Computer Information & Security I

Project Specification 2026

Release Date: 19 February 2026

Submission Date: 19 May 2026

Lecturer: Mr S.M. Makura

Total: 100 Marks

Instructions

1. This project must be completed on an individual basis. **No group work is allowed.** It must be the **student's own work.** You must clearly indicate if you have objects / components coming from other resources. You must clearly show the uniqueness of your solution in comparison to other existing solutions.
2. An anti-plagiarism approach will be strictly enforced. Please familiarise yourself with the UP-Plagiarism Prevention policy. All plagiarism cases pertaining to any aspect of the project will be handled over to UP Legal for further investigation. You will have to submit a signed form acknowledging that you have read and are in agreement with the **UP Plagiarism Prevention policy which can be read here:**
https://www.up.ac.za/media/shared/1/ZP_Files/s5106-19-plagiarism-prevention-policy.zp181077.pdf).
3. Online submission of project supporting documentation:
 - Submit the following:
 - 1 page **research overview (in your own words)** that relates to your solution. Include literature references.
 - A detailed bibliography (separate from the 1 page above). (This must be your detailed bibliography compiled by yourself).
 - 1 to 2 page requirements specification of your **OWN** unique solution

- **UML design diagrams** of your OWN unique solution.
- A Turnitin report for your submissions.
- Your submissions must be in Word or PDF (doc .docx .pdf file).
- All design diagrams must be in formal UML notation.

NO LATE submissions will be accepted after the submission date has lapsed regardless of whatever reason you give.

Project Topic: AI-Powered Insider Threat Detection System

Objective:

Develop a lightweight AI-powered prototype system that detects potential insider threats in a corporate environment by analysing user behavioural features from organisational logs.

The system must classify user activity as:

- Malicious Insider Activity
- Normal / Benign Behaviour

The system must also provide an explanation of why a specific behaviour was flagged as suspicious.

Project Tasks:

1. Research Overview (part of the documentation submitted at the end):

You must include:

- A 1 page summary explaining what insider threats are and why they pose a major risk to organisations.
- A brief explanation at most of the page of how Artificial Intelligence and Machine Learning can be applied to behavioural threat detection.
- A comparison between traditional rule-based monitoring and AI-driven detection approaches.

You must also research and summarise the following published paper:

Shoderu, G., Baror, S.O., Makura, S.M. and Modupe, A., 2025. Digital Forensic Readiness to Mitigate Insider Threats in the SaaS Cloud Environment. The Indonesian Journal of Computer Science, 14(6). You can download the paper here:

<http://ijcs.net/ijcs/index.php/ijcs/article/view/5059>

You must:

- Clearly explain **how the paper's findings relate to your solution**
- Identify at least **one gap in the existing literature** that your project attempts to address.

This must be written in **your own words**.

2. Dataset Preparation:

You use the following Kaggle dataset:

*Insider Threat Dataset for Behavioral Features & Threat Classification
Dataset for Security Assessment*

Available on **Kaggle**

Dataset link: <https://www.kaggle.com/datasets/ahmeduzaki/insider-threat-dataset-for-corporate-environments>

The use of alternative datasets **is not permitted**, in order to ensure fairness and consistency during evaluation.

- Download the prescribed dataset from Kaggle.
- Perform data cleaning and preprocessing.
- Handle missing values and encode categorical variables where applicable.
- Clearly describe:
 - Which features you selected
 - Why those features are relevant to insider threat detection

You must demonstrate understanding of behavioural indicators such as:

- Login time patterns
- File access activity
- Privilege usage
- Abnormal access frequency
- Data transfer behaviour

3. Model Development:

- Implement **at least one** machine learning model (e.g., Logistic Regression, Random Forest, Decision Tree, Support Vector Machine etc).
- Train the model using the provided dataset.
- Evaluate performance using:
 - Accuracy
 - Precision
 - Recall
 - F1-Score
 - Confusion Matrix

You must justify:

- Why you selected your model
- Why it is appropriate for insider threat detection

4. **Prototype Implementation:**

Develop a simple prototype system (desktop or web-based) that allows:

- Uploading of a behavioural data record (CSV input), or
- Selection of a sample employee behaviour profile.

The system must output:

- Classification result (Malicious / Normal)
- Confidence score
- A short explanation of key risk indicators contributing to the decision

You are welcome to use **any programming language** you are comfortable with.

5. **AI Explainability:**

Your prototype must include an explainability feature, such as:

- Feature importance ranking
- Highlighting abnormal behavioural indicators
- Risk score breakdown
- Simple human-readable explanation of why the activity was flagged

The explanation must be understandable to a non-technical manager.

6. **Testing Scenarios:**

You must test the system using multiple records from the dataset.

Document:

- True positives
- False positives
- False negatives

Analyse cases where the system performs poorly and suggest realistic improvements.

7. Documentation:

Include:

- A 1-page explanation of the model selection process.
- UML diagrams including:
 - System architecture diagram
 - Activity diagram or sequence diagram
- A discussion of:
 - Limitations of behavioural AI detection
 - Ethical considerations (e.g., employee privacy, monitoring risks)

All diagrams must use formal UML notation.

Submission Requirements:

1. Codebase

- Well-documented source code.
- Clear instructions on how to deploy and to run the prototype.

2. AI Model

- Submit trained model file (if applicable).

3. Testing Report

- Provide evaluation metrics.
- Include screenshots or evidence from the prototype.

4. Documentation

- Research overview.
- UML diagrams.
- Requirements specification.
- Bibliography.
- Turnitin report.

Evaluation Criteria:

1. Model Performance

- Accuracy and effectiveness of insider threat detection.

2. Prototype Functionality

- Usability of the system.
- Clarity of classification output.
- Quality of explainability.

3. Documentation

- Clarity and depth of research.
- Logical reasoning behind feature and model selection.
- Correct and professional UML design.

The documentation for the project counts for the marks for the project as well as the project demonstrations. The demonstrations will be done either physical face-to-face hands-on demonstrations to the lecturer or using an online demonstration.

Failure to demo, means that you will fail the project (non-negotiable). Students should therefore ensure that they demo their projects as well ensure that their documentation includes a well- structured design of their solution (it must be in UML notation). The due date for the project assignment is **19 May 2026**. No late submissions or extensions allowed.

Happy coding 😊